Final Examination Practice Sheet - ANSWER KEY

Course Code: INT396
Course Title: UNSUPERVISED LEARNING
Section(s): K3P24MK, K3P24WV
Syllabus Covered: Unit 1 to Unit 6
Pattern: per unit, 8 MCQs (1 mark) and 2 long questions (10 marks)
Purpose: self-practice for the end-term examination

For the instructor. Every question is a reworded version of a question in the course question bank (scenario, names or numbers changed; numerical answers recomputed). The "Bank" column gives the source question number in that unit.

Unit 1: Foundations of Unsupervised Learning

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(B)**	Manhattan distance. Manhattan distance sums absolute coordinate differences.	Q6
2	**(D)**	A label column giving the correct answer for each record. Unsupervised data has no target / label column.	Q8
3	**(A)**	0 to 1. With non-negative features the angle is at most 90 degrees, so cosine lies in 0 to 1.	Q12
4	**(C)**	They spend in the same proportions across categories, even if their totals differ. Cosine 1.0 means the vectors point in the same direction.	Q17
5	**(C)**	Anomaly detection. Finding the rare records unlike the bulk is anomaly detection.	Q20
6	**(A)**	Income dominates the distance. The feature with the far larger numeric range dominates.	Q18
7	**(D)**	There are no true labels to compare the clusters against. Accuracy needs a ground truth, which unlabelled data lacks.	Q13
8	**(B)**	Semi-supervised learning. A little labelled data plus much unlabelled data is semi-supervised.	Q15

Part B - Expected answers and marking scheme

9. A video-streaming service represents each user as a vector of minutes watched in each genre. *(10 Marks) [Bank Q3]*

(a) Explain what Euclidean distance, Manhattan distance and cosine similarity each measure about a pair of users.

Euclidean: straight-line distance, square root of the sum of squared differences; large single gaps dominate. Manhattan: sum of absolute differences per genre; every gap counts linearly. Cosine: the cosine of the angle between the two vectors; it compares the pattern (direction), not the size. (6 marks.)

(b) Two users watch the same mix of genres, but one watches three times as many minutes in total. State which of the three measures treats them as most alike, and give the reason.

Cosine similarity. The two vectors point in the same direction (one is three times the other), so cosine = 1, while Euclidean and Manhattan distances are large because the totals differ. (4 marks.)

10. Distinguish supervised, unsupervised and semi-supervised learning, using an e-commerce company as the setting. *(10 Marks) [Bank Q2]*

(a) For each of the three, describe the data it requires and the question it answers.

Supervised: labelled records (features plus a known target); predicts the target for new records. Unsupervised: unlabelled records only; finds hidden structure such as groups or anomalies. Semi-supervised: a few labelled and many unlabelled records; uses the unlabelled data to improve a predictor built from the few labels. (6 marks.)

(b) Give one e-commerce application of each.

Supervised: predicting whether an order will be returned from past labelled returns. Unsupervised: segmenting customers by browsing and purchase behaviour. Semi-supervised: classifying product reviews as genuine or fake when only a few reviews have been checked by staff. (4 marks.)

Unit 2: Partition-Based Clustering

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(C)**	(4, 7). Mean x = 12/3 = 4, mean y = 21/3 = 7.	Q15

2	**(A)**	Centroid 10, contribution 72. Centroid 10; (6^2 + 0 + 6^2) = 72.	Q16
3	**(B)**	8, with cost 17. Costs: 3 gives 22, 8 gives 17, 20 gives 29; a medoid must be an actual record.	Q17
4	**(D)**	0.64. Probability is proportional to squared distance: 16 / (9 + 16) = 0.64.	Q18
5	**(A)**	-1.5. z = (44 - 50) / 4 = -1.5.	Q19
6	**(C)**	k = 3. The drop collapses from 350 (k 2 to 3) to 20 (k 3 to 4).	Q23
7	**(D)**	It is pulled towards the mansion, because the centroid is a mean. A mean is sensitive to extreme values.	Q24
8	**(B)**	k-Means++. k-Means++ spreads the starting centres far apart.	Q11

Part B - Expected answers and marking scheme

9. A gym runs k-Means on its members using the number of visits per month (0 to 30) and the annual membership fee in rupees (5,000 to 60,000), without scaling either column. *(10 Marks) [Bank Q1]*

(a) Analyze which of the two features decides the clusters, and give the reason.

The fee decides the clusters. k-Means uses Euclidean distance, and the fee varies over tens of thousands while visits vary over at most 30, so differences in fee swamp differences in visits; the clusters become fee bands and visit behaviour is effectively ignored. (5 marks.)

(b) Infer how the clusters would change after both columns are standardised.

After standardisation both features have mean 0 and standard deviation 1, so each contributes comparably to the distance. The clusters would then reflect combinations of visit frequency and fee (for example high-fee low-visit members), which are more useful for retention decisions. (5 marks.)

10. A retailer runs k-Means for k = 2 to 7. The inertia falls steeply up to k = 4 and then flattens, and the silhouette score is highest at k = 4. *(10 Marks) [Bank Q4]*

(a) Analyze what inertia and the silhouette score each tell you about a clustering.

Inertia is the within-cluster sum of squared distances to the centroids: it measures compactness only. The silhouette compares, for each point, the mean distance to its own cluster (a) with the mean distance to the nearest other cluster (b): s = (b - a) / max(a, b); it combines compactness and separation, ranging from -1 to +1. (5 marks.)

(b) Infer the value of k you would choose, give the reason, and explain why inertia alone cannot be used to pick k.

Choose k = 4: it is the elbow of the inertia curve and the silhouette peak, so both criteria agree. Inertia alone always decreases as k grows (reaching zero when every point is its own cluster), so its minimum cannot choose k. (5 marks.)

Unit 3: Hierarchical and Density-Based Clustering

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(D)**	eps. eps is the neighbourhood radius.	Q5
2	**(B)**	It is not a core point; it may still be a border point if it lies within eps of a core point. It has fewer than 5 points in its neighbourhood, so it is not core.	Q13
3	**(A)**	19. Each merge removes one cluster: 20 - 1 = 19.	Q14
4	**(C)**	11. Complete linkage takes the maximum.	Q15
5	**(B)**	6. 48 / (2 x 4) = 6.	Q17
6	**(D)**	eps is too small or MinPts is too large. Few points reach MinPts neighbours, so few core points exist.	Q12
7	**(C)**	Single linkage. Single linkage uses the closest pair.	Q11
8	**(A)**	It needs all pairwise distances, so cost grows with the square of the number of records. The n x n distance matrix is the bottleneck.	Q24

Part B - Expected answers and marking scheme

9. An analyst runs k-Means with k = 2 on GPS points that form two crescent-shaped (half-moon) groups, and each resulting cluster cuts across both crescents. *(10 Marks) [Bank Q4]*

(a) Explain why k-Means cannot find these two groups.

k-Means assigns each point to the nearest centroid, so its clusters are convex regions separated by straight boundaries. A crescent is not convex: its ends are closer to the other crescent's centroid than to their own, so the straight boundary slices through both groups. It also minimises squared distance, which favours round, equal-sized clusters. (5 marks.)

(b) State which method you would use instead, and what that method needs to be set before it runs.

DBSCAN (or single-linkage hierarchical clustering). DBSCAN follows connected dense regions of any shape. It needs eps (the neighbourhood radius, chosen for example from a k-distance plot) and MinPts (minimum points for a dense region); the features should also be on a comparable scale. (5 marks.)

10. Compare single linkage, complete linkage and average linkage for clustering shop locations along a highway. *(10 Marks)* *[Bank Q2]*

(a) State how each of the three measures the distance between two clusters.

Single: distance between the two closest members, one from each cluster. Complete: distance between the two furthest members. Average: mean of all cross-cluster pairwise distances. (5 marks.)

(b) State which one gives long chain-like clusters and which gives compact ones, with the reason, and which is most affected by outliers.

Single linkage gives long chains: two groups merge as soon as any one pair is close, so a line of shops along the highway links up. Complete linkage gives compact clusters because a merge needs even the furthest pair to be close. Single linkage is most affected by outliers and noise points, which can bridge separate groups (chaining); complete linkage is also pulled by a single far point. (5 marks.)

Unit 4: Dimensionality Reduction and Representation Learning

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(A)**	3. 45 + 25 + 15 = 85 per cent at three components.	Q13
2	**(C)**	75 per cent. (120 - 30) / 120 = 75 per cent.	Q14
3	**(B)**	15 per cent. 80 - 65 = 15 per cent.	Q15
4	**(D)**	The direction of greatest variance. PC1 is the direction of maximum variance.	Q6
5	**(C)**	Weight dominates the components because of its larger raw variance. PCA maximises variance, which depends on the unit.	Q16
6	**(A)**	Little: t-SNE preserves local neighbourhoods, not large distances. t-SNE distances between clusters are not meaningful.	Q18
7	**(D)**	The reconstruction error. An autoencoder is trained to reproduce its input.	Q24
8	**(B)**	PCA, since each component is an explicit weighted sum of the original features. PCA loadings are interpretable; the others are non-linear.	Q21

Part B - Expected answers and marking scheme

9. Assess the need for dimensionality reduction on a bank dataset holding 10,000 customers described by 1,200 features. *(10 Marks)* *[Bank Q3]*

(a) Evaluate two benefits it could deliver and one risk it introduces.

Benefits (any two): reduces the curse of dimensionality so distances become meaningful again; removes redundant or correlated features and noise; cuts storage and computation; allows visualisation. Risk: information carried by low-variance components (possibly the signal of interest, such as fraud) may be discarded, and the new features are harder to interpret. (5 marks.)

(b) Judge whether the features must be standardised before PCA is applied, explaining your reasoning.

Yes, if the features are in different units or ranges (balances in rupees, counts, ratios). PCA picks directions of maximum variance, so un-standardised large-scale features would dominate purely because of their unit. Standardising to zero mean and unit variance makes each feature contribute on an equal footing. If all features already share one unit and scale, standardisation is less critical. (5 marks.)

10. Critique the claim in a project report: 'Our autoencoder compresses 300 sensor features into 15, and since the reconstruction error is very low, the 15 features contain all the useful information.' *(10 Marks)* *[Bank Q4]*

(a) Assess what a low reconstruction error does and does not establish.

It establishes that the 15-dimensional code can rebuild the inputs closely on the data measured (possibly only on training data). It does not show that the code is useful for the task, that rare but important patterns are kept (they contribute little to average error), that it generalises to new data, or that the network has not simply overfitted. (5 marks.)

(b) Recommend the evidence the report should have presented instead.

Reconstruction error on held-out test data; comparison with a PCA baseline at the same dimension; performance of a downstream task (clustering or prediction) using the 15 features against the original 300; per-feature or per-class error to check rare patterns; stability across training runs. (5 marks.)

Unit 5: Association Rule Mining and Anomaly Detection

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(B)**	0.15. 120 / 800 = 0.15.	Q13
2	**(D)**	2.0. Lift = 0.10 / (0.25 x 0.20) = 2.0.	Q15
3	**(A)**	The consequent is common anyway, so the rule reveals little. Lift near 1 means near	Q11

		independence.	
4	**(C)**	If an itemset is infrequent, every larger itemset containing it is also infrequent. This is the downward-closure (Apriori) property.	Q8
5	**(D)**	Isolation Forest. Anomalies have short isolation paths.	Q6
6	**(B)**	Its local density with the local densities of its neighbours. LOF is a relative density measure.	Q10
7	**(A)**	A missed interaction (false negative), because the patient is harmed. Missing a real interaction risks patient harm.	Q23
8	**(C)**	Most itemsets are now excluded, and some genuine but rarer patterns may be lost. A higher threshold keeps only very common itemsets.	Q16

Part B - Expected answers and marking scheme

9. A bookshop mined association rules from 2,000 bills. The rule {notebook} -> {pen} has support 0.18, confidence 0.90 and lift 1.05. *(10 Marks) [Bank Q1]*

(a) State what support, confidence and lift each measure about a rule.

Support: fraction of all bills containing both notebook and pen (how common the rule is). Confidence: P(pen | notebook), the fraction of notebook bills that also contain a pen (how reliable). Lift: confidence divided by the support of pen, i.e. how much more often they occur together than if independent. (5 marks.)

(b) Explain why this rule, despite its high confidence, may be of little use, and state what a lift close to one means.

Pens are bought in about 0.90 / 1.05 = 86 per cent of all bills anyway, so a notebook buyer is barely more likely to buy a pen than any other customer; the rule adds almost nothing for promotion or placement. Lift close to 1 means the two items are close to independent; above 1 positive association, below 1 negative. (5 marks.)

10. A government audit office must flag suspicious claims among 2 million health-insurance claims, and must explain every flag to its investigators. *(10 Marks) [Bank Q3]*

(a) Describe how Isolation Forest and Local Outlier Factor each decide that a claim is unusual.

Isolation Forest builds many random trees that split on random features at random values; anomalies are isolated in few splits, so a short average path length gives a high anomaly score. LOF estimates each record's local density from its k nearest neighbours and compares it with the neighbours' densities; a much lower density than its neighbours gives a high LOF score. (5 marks.)

(b) Recommend one of the two on the basis of speed and explainability, with your reason.

Isolation Forest: it scales nearly linearly with the number of records and does not need all nearest-neighbour distances, so it handles 2 million claims quickly; feature-split paths give a usable explanation. LOF needs k-nearest-neighbour searches over all records (roughly quadratic cost) and its density ratio is harder to explain to investigators. LOF may still be preferred for small data with clusters of very different density. (5 marks.)

Unit 6: Evaluation and Applications of Unsupervised Learning

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(C)**	0.67. s = (9 - 3) / 9 = 0.67.	Q15
2	**(A)**	-0.25; it is probably in the wrong cluster. s = (6 - 8) / 8 = -0.25.	Q16
3	**(D)**	Because the silhouette always lies between -1 and +1. s = (b - a) / max(a, b) cannot exceed 1 in size.	Q7
4	**(B)**	It lies on the boundary between two clusters. a and b are nearly equal.	Q9
5	**(B)**	Internal versus external validation measures. Silhouette is internal; ARI is external.	Q8
6	**(C)**	The structure is unstable and should not be acted upon yet. Stability across resamples is required.	Q18
7	**(A)**	Reproduce the same bias in its groups. Features carry the history they were drawn from.	Q13
8	**(D)**	X's clusters are more compact, but the silhouette, which also counts separation, favours Y. Inertia measures compactness only.	Q20

Part B - Expected answers and marking scheme

9. Apply an unsupervised learning workflow to the following brief: a food-delivery company holds a year of anonymised orders (time, order value, cuisine, delivery area) and wishes to understand the kinds of customer it serves. *(10 Marks) [Bank Q1]*

(a) State the preprocessing required and the features you would construct.

Clean duplicates and cancelled orders; handle missing values; aggregate orders to one row per customer. Features: order frequency, average and total spend, share of orders by cuisine, share by time of day / weekday vs weekend, recency, number of distinct areas. Encode categories as proportions, treat outliers (bulk corporate orders), and standardise. (5 marks.)

(b) State the method you would use and how you would validate the result.

k-Means (or a Gaussian mixture) after optional PCA; choose k with the elbow and silhouette. Validate with internal measures (silhouette, Davies-Bouldin), stability across resamples or seeds, and business review: profile each segment (e.g. late-night students, weekend families) and check it is interpretable and actionable. (5 marks.)

10. A telecom company wants to act on a customer segmentation and must first establish that it is stable. *(10 Marks) [Bank Q4]*

(a) Describe a procedure for testing stability by resampling or by perturbing the data.

Draw many bootstrap samples or random 80 per cent subsamples (or add small noise / change the random seed); re-run the same clustering with the same k on each; map each new clustering to the original and measure agreement with the Adjusted Rand Index or the Jaccard similarity of matched clusters; also repeat for neighbouring k. (5 marks.)

(b) Explain what result would show that the segments are genuine structure rather than an artefact of one particular sample.

High and consistent agreement (e.g. ARI above about 0.8), similar segment sizes and centroids across runs, and the same customers repeatedly grouped together indicate genuine structure. Low or highly variable agreement, or segments that appear and disappear, indicate an artefact of the sample or the initialisation. (5 marks.)

----- End of Answer Key -----